

Inventories and Monetary Policy

Marco Brianti* Vito Cormun[†] Jianlin Wang[‡]

June 27, 2026

Abstract

We show that the transmission of monetary policy depends sharply on the inventory position of firms. We document that a monetary policy shock affects output and prices more than twice as much when firms hold low inventories. In the high-inventory regime the same shock produces responses that are small and often statistically insignificant. The result holds at the aggregate, sectoral, and firm levels. We rationalize it in a New Keynesian model with stock-elastic demand and sticky prices. The model highlights a new channel: abundant inventories raise the effective degree of price stickiness, as firms carrying more finished goods into the future become more reluctant to change prices today.

JEL classification: E22, E31, E32, E52

Keywords: Inventories, Monetary policy transmission, state dependence, inventory-to-sales ratio, threshold VAR, local projections, price stickiness

*University of Bologna, marco.brianti@unibo.it

[†]Santa Clara University, vcormun@scu.edu

[‡]Santa Clara University, jianlin.wang@scu.edu

1 Introduction

Inventories are among the most volatile components of aggregate spending, yet their role in monetary policy transmission has received little systematic attention. This paper argues that the oversight is consequential: the inventory position of firms is a first-order determinant of how effectively monetary policy shifts output and prices.

Our main finding is that monetary policy is more than twice as potent when inventories are scarce. Following a 25-basis-point increase in the one-year Treasury rate, industrial production declines by more than 1% at peak, the CPI by roughly 0.4%, and commodity prices by roughly 1% when firms enter the episode with low inventories. In the high-inventory regime, the same shock elicits responses that are small and statistically insignificant throughout. We establish this asymmetry at three levels of aggregation. At the aggregate level, we estimate a Bayesian threshold VAR identified with external instruments, exploiting time-series variation in the business-sector inventory-to-sales ratio to classify the economy into high- and low-inventory regimes. At the sector level, we corroborate the finding using a panel of manufacturing and wholesale industries, estimating state-dependent local projections that interact the monetary policy shock with a sector-specific inventory indicator. At the firm level, we provide further evidence using state-dependent local projections that include sector-time fixed effects, directly absorbing previously documented state-dependent channels of monetary policy—such as financial conditions—and confirming that the inventory channel operates independently within narrowly defined sectors. The asymmetric amplification result is robust across a wide range of choices for the filter, instrument, sample, and variable specification.

These findings shed new light on a long-standing puzzle. Standard production-smoothing models predict that firms use inventories to buffer demand fluctuations, insulating output from shocks (Blinder, 1982). The data, however, consistently reject this prediction: production is more volatile than sales, and the inventory-to-sales ratio moves countercyclically (Blinder and Maccini, 1991). Confronted with this failure, the literature has largely con-

cluded that inventories play a negligible role in the aggregate business cycle (Khan and Thomas, 2007), sparing macroeconomists from an additional state variable. Our analysis reframes this conclusion: inventories may not amplify unconditional fluctuations, but they profoundly shape the economy’s response to monetary policy.

To rationalize this finding, we embed inventories into an otherwise standard New Keynesian model following Lubik and Teo (2012). The model has two key departures from the textbook framework. First, a firm’s demand is increasing in its own stock of finished goods as in Bils and Kahn (2000), capturing the idea that firms with low production stock don’t have to turn away their customers or that larger amounts of readily available goods reduce search costs in finding buyers (Den Haan, 2014). Second, convex price adjustment costs are proportional to the volume of goods sold. These two features together imply that high beginning-of-the-period inventories raise effective price stickiness. When a firm holds more inventory, it carries more stock into the next period and expects higher future sales—since its own stock raises demand for its goods. With higher expected future sales, firm’s current pricing decision become relatively more forward-looking. As a result, firms are more reluctant to change current prices, and the inflation response to a monetary shock is relatively muted. Conversely, when inventories are low, expected future sales are smaller, future price adjustment is cheaper, and firms respond more freely to the shock. An intertemporal Euler equation links current marginal cost to expected future costs through the inventory stock, adding a further source of state dependence in cost dynamics. We show that the model produces qualitatively the right asymmetry in both inflation and output responses across inventory states, and discuss its limitations as motivation for future theoretical work.

Our paper contributes to two strands of the literature. The first concerns inventories and monetary policy transmission. An early literature documents that inventory dynamics are a key conduit for monetary policy: Blinder and Maccini (1991) show that inventory investment declines account for 87% of the average peak-to-trough drop in GNP across postwar

recessions; [Bernanke and Gertler \(1995\)](#) find that inventory disinvestment is among the earliest and sharpest responses to a monetary tightening; and [Kashyap et al. \(1994\)](#) show in the cross-section that firms lacking access to public capital markets cut inventories far more sharply during episodes of tight money. [de Sousa Rodrigues and Willems \(2024\)](#) provide a closely related contemporary contribution, documenting a cost-of-carry channel: rising interest rates raise the cost of holding inventories, so firms with larger stocks cut prices to liquidate, making monetary policy more effective at reducing goods prices when inventory levels are elevated. Our paper identifies a complementary mechanism operating in the opposite direction: when inventories are *low*, neither output nor prices can be buffered, generating larger contractions in both. Together, the two channels imply a non-linear, state-dependent relationship between the inventory-to-sales ratio and monetary policy effectiveness.

The second strand concerns state-dependent supply and pricing responses. [Boehm and Pandalai-Nayar \(2022\)](#) document convex aggregate supply curves, with price responses to demand shocks that are markedly larger when capacity is tight; the industry capacity utilization rate is a sufficient statistic for the supply elasticity. [Ahlander et al. \(2026\)](#) confirm this at the firm level using Swedish producer-price micro data: capacity-constrained firms raise prices substantially more in response to an expansionary monetary shock, with the difference widening when competitors are similarly constrained. [Bai et al. \(2026\)](#) apply a similar logic to global supply chain disruptions: using a structural threshold VAR and local projections, they find that during disruption episodes — when logistical bottlenecks steepen the aggregate supply curve — contractionary monetary policy produces a larger price decline at a smaller output cost, easing the stabilization trade-off. [Mehrotra et al. \(2025\)](#) embed the inventory-to-sales ratio directly in a New Keynesian Phillips curve and show that it helps explain both the missing disinflation of 2009–2011 and the COVID-era inflation surge through time-varying retail markups. Our contribution to this strand is to show that the inventory-to-sales ratio governs not just pricing behavior but the en-

tire macroeconomic transmission of monetary policy shocks, amplifying quantity responses alongside price responses.¹

The rest of the paper proceeds as follows. Section 2 presents the empirical evidence, beginning with the aggregate threshold VAR in Section 2.1, continuing with the sector-level local projections in Section 2.2, and concluding with the firm-level local projections in Section 2.3. Section 3 develops the theoretical model. Section 4 concludes.

2 Empirics

2.1 Aggregate evidence

2.1.1 General setting

There are several possible approaches to estimating the state-dependent effects of a monetary policy shock as a function of the amount of inventories. As a first step, we follow [Alessandri and Mumtaz \(2017, 2019\)](#) and estimate a Bayesian threshold VAR. We start with this model because it allows us to estimate the state-dependent effects of a monetary policy shock through generalized impulse response functions (GIRFs). As argued by [Francis et al. \(2023\)](#) and [Gonçalves et al. \(2024\)](#), GIRFs account for endogenous transitions across states induced by the shock, thereby avoiding the potential bias of local projections (LPs), which instead capture average effects across regimes. Moreover, the reduced-form VAR coefficients are estimated in a single step, improving overall model fit.

Reduced-form VAR Consider the following threshold VAR model:

$$Y_t = \left[c_1 + \sum_{j=1}^P B_{1,j} Y_{t-j} + u_{1,t} \right] S_t + \left[c_2 + \sum_{j=1}^P B_{2,j} Y_{t-j} + u_{2,t} \right] (1 - S_t) \quad (1)$$

where $Y_t = [y_{1,t}, y_{2,t}, \dots, y_{n,t}]'$ is a $n \times 1$ vector of endogenous variables; $u_{i,t} \sim \mathcal{N}(0, \Sigma_i)$ is a $n \times 1$ vector of reduced-form innovations in regime $i \in \{1, 2\}$, with Σ_i the associated

¹ Our paper also relates to the broader literature on state-dependent monetary policy transmission across financial conditions (e.g., [Alessandri and Mumtaz, 2017, 2019](#)). We complement this literature by identifying a real-side amplifier — the inventory position — that is distinct from financial frictions and operates through firms' ability to buffer demand, rather than their access to external finance.

$n \times n$ covariance matrix; and c_i and $B_{i,1}, \dots, B_{i,p}$ denote, respectively, the $n \times 1$ vector of intercepts and the $n \times n$ matrices of autoregressive slopes in regime $i = 1, 2$. The regime indicator S_t is defined as:

$$S_t = 1 \iff y_{n,t-d} \leq y_n^*, \quad (2)$$

where $y_{n,t}$ is the threshold variable, assumed without loss of generality to be the last element of Y_t , y_n^* is the threshold value, and $d \geq 1$ is the delay parameter governing the minimum lag with which $y_{n,t}$ can affect regime classification. Accordingly, $S_t = 1$ identifies the “low” regime, prevailing when the threshold variable lies below y_n^* , while $S_t = 0$ identifies the “high” regime.

Estimation The unknown parameters $\theta \equiv \{B_{i,j}, c_i, \Sigma_i\}_{i=1,2; j=1,\dots,p}$, together with the threshold value y_n^* and the delay parameter d , are estimated by Bayesian methods. We adopt a Minnesota prior, which shrinks the VAR slope matrices toward univariate AR(1) processes and toward zero at longer lags, complemented by a sum-of-coefficients prior that further pulls the sum of lag coefficients toward a unit root. Importantly, both priors are applied symmetrically across regimes, implying a conservative approach: the two states share a common prior that shrinks their coefficients toward being identical, so that differences across regimes emerge only to the extent warranted by the data.

The joint posterior distribution does not admit a closed form and is simulated via a Markov chain Monte Carlo (MCMC) algorithm that combines a Gibbs sampler for θ with a Metropolis-Hastings step for the threshold y_n^* and a discrete Gibbs step for the delay d . Further details on the estimation procedure are provided in Online Appendix C.1.

Baseline specification The VAR is estimated at a monthly frequency and includes six variables. The 1-year Treasury bill rate, in basis points, is used to normalize the impulse responses to a monetary policy shock. The industrial production index (IP) and the producer price index (PPI), both in logs, capture real activity and nominal dynamics, respec-

tively.² The log-transformation of commodity prices and the excess bond premium (EBP, in basis points), both highly forward-looking, serve as controls for financial conditions. The last variable, which serves as the threshold variable, is the inventory-to-sales ratio, whose specification is discussed below.

The numerator of the threshold variable is finished-goods inventories of the business sector. We focus on output rather than input inventories because finished-goods stocks directly measure firms’ capacity to fulfill demand.³ The denominator is the cumulative sales over the current and preceding two months, approximating the last quarter’s total sales rather than current-month sales alone. This allows for a slow-moving normalization of the available stock of inventories, so that variations in the ratio are primarily driven by changes in the numerator rather than by high-frequency swings in sales that partly reflect variation in aggregate demand. Moreover, since the inventory-to-sales ratio exhibits pronounced low-frequency trends, we apply a high-pass filter that removes fluctuations with frequencies lower than ten years. Details on data sources are provided in Appendix A.

Finally, the VAR is estimated with $p = 12$ lags, consistent with the monthly frequency of the data and the sample runs from December 1981 — the earliest date at which business-sector final-goods inventories are jointly available across manufacturing, wholesale, and retail — to January 2020, which excludes the COVID-19 period.

Regime classification Figure 1 displays the estimated regime classification. The black line is the filtered inventory-to-sales ratio evaluated at the posterior mode of the delay parameter d , the red horizontal line is the posterior median of the estimated threshold y_n^* , and the shaded vertical areas identify periods in the low-inventory regime when using the posterior mode for d and the posterior median for y_n^* .

The estimated threshold lies slightly above zero, reflecting the fact that the posterior does not simply coincide with the average of the state variable $y_{t,n}$ — which is zero by the

² In the baseline we use the PPI rather than the CPI or other price measures, to ease comparison with the sectoral analysis in Section 2.2.

³ A firm may hold large input inventories yet remain unable to supply if even one key input is missing, whereas final-goods inventories unambiguously signal that finished goods are available for sale.

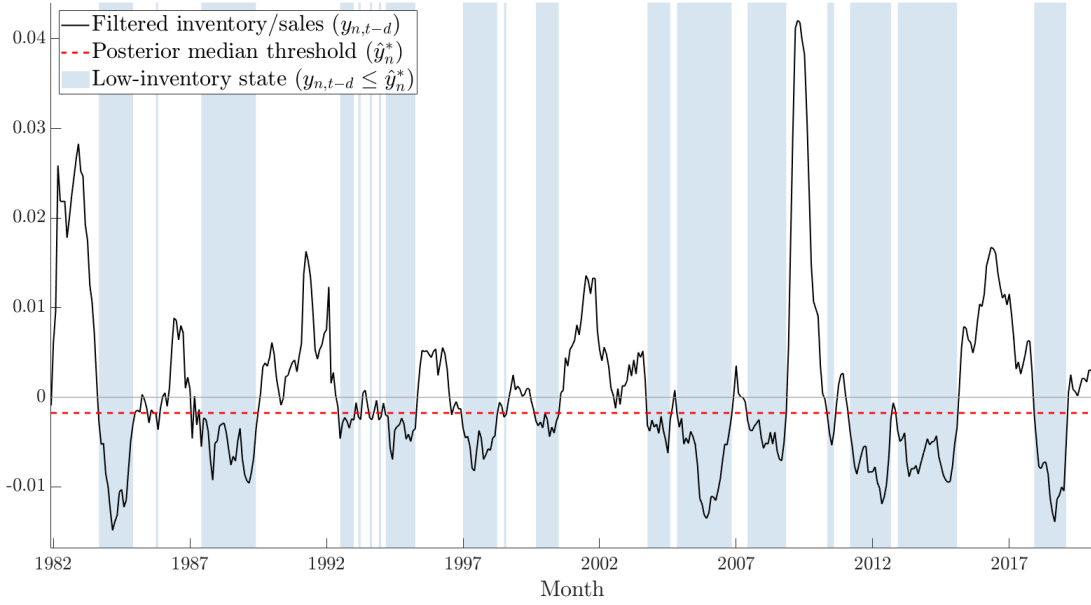


Figure 1: Filtered inventory-to-sales ratio and regime classification

Note: The black line is the cyclical component of the finished-goods inventory-to-sales ratio of the business sector, obtained by applying a high-pass filter that removes fluctuations with frequencies lower than 120 months, and evaluated at the posterior mode of the delay parameter d . The red horizontal line is the posterior median of the estimated threshold y_n^* . Shaded areas identify periods classified as the low-inventory regime ($S_t = 1$), defined using the posterior mode of d and the posterior median of y_n^* to assign each observation to a regime. Sample: December 1981 – January 2020. See Online Appendix C.1 for a detailed description of the estimation of the reduced-form VAR.

property of the filter — but is also shaped by the likelihood of the data. Moreover, the regime classification does not simply track the business cycle: while some low-inventory episodes coincide with recessions, the figure reveals variation that is independent of aggregate downturns, suggesting that inventory positions may contain information about monetary policy transmission beyond what is captured by standard measures of economic slack.

Structural identification Following [Stock and Watson \(2018\)](#), we identify the causal effect of monetary policy by augmenting the reduced-form threshold VAR with an external instrument, using existing off-the-shelf proxies for the monetary policy shock series. The key additional step relative to a standard external-instrument VAR is that the first-stage regression is estimated separately for each regime. Specifically, let $\hat{u}_{i,t}$ denote the estimated

vector of reduced-form residuals in regime $i \in \{1, 2\}$. For each state, we regress $\hat{u}_{i,t}$ on the proxy for the monetary policy shock $\hat{\varepsilon}_t$:

$$\hat{u}_{i,t} = \gamma_i \hat{\varepsilon}_t + v_{i,t}, \quad i = 1, 2, \quad (3)$$

where under conventional assumptions the estimated coefficient vector $\hat{\gamma}_i \in \mathbb{R}^n$ captures the impact effect of the monetary policy shock on the endogenous variables Y_t in regime i . The baseline proxy is the monetary policy surprise series of [Bauer and Swanson \(2023\)](#), which purges high-frequency interest rate surprises of their predictable component using publicly available macroeconomic information, and represents one of the most recent advances in the measurement of monetary policy shocks. As discussed in Section [2.1.3](#), the results are robust to the use of alternative monetary policy proxies.

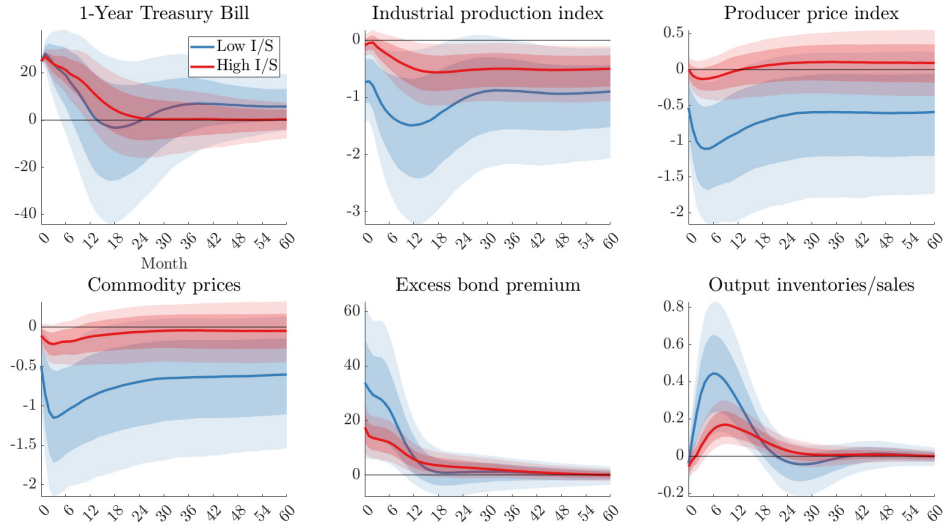
This approach offers two key advantages. First, the reduced-form stage and the structural stage are independent and can be estimated separately. This separation property is particularly valuable when studying monetary policy transmission, since monetary policy shock proxies are available only over a restricted time window relative to the availability of standard macroeconomic indicators. Thanks to this separation, we can estimate the reduced-form VAR parameters over the full sample for which all endogenous variables are available, and then identify the state-specific impact vectors $\hat{\gamma}_i$ on the shorter overlapping sample between the reduced-form residuals and the monetary policy proxy $\hat{\varepsilon}_t$. Second, the threshold VAR framework allows us to construct state-dependent generalized impulse response functions (GIRFs; [Koop et al. 1996](#)). State-dependent GIRFs condition on the regime prevailing at the time the shock occurs and allow the economy to transition between regimes endogenously after impact. As argued by [Francis et al. \(2023\)](#) and [Gonçalves et al. \(2024\)](#), this avoids the potential bias of local projections, which instead capture average effects across regimes. We describe the construction of the GIRFs in Online Appendix [C.2](#).

2.1.2 Results

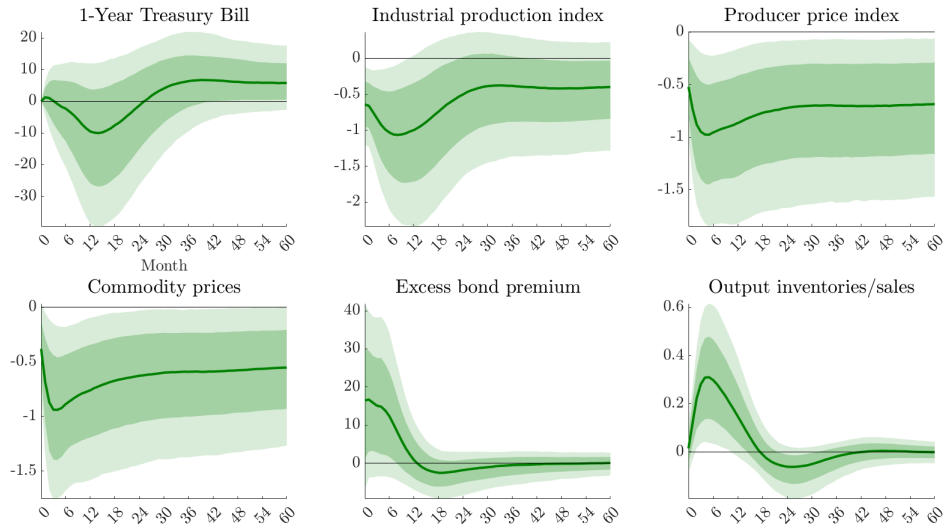
Figure 2 presents the GIRFs to a contractionary monetary policy shock for all variables in the VAR (Panel a) together with the corresponding differences across regimes (Panel b). The shock is normalized so that the impact effect on the one-year Treasury bill equals 25 basis points. Solid lines are posterior medians; dark and light shaded areas correspond to 68% and 90% posterior credible bands, respectively. Blue (red) lines and bands correspond to the low- (high-) inventory regime, defined as periods when the inventory-to-sales ratio lies below (above) the estimated threshold y_n^* .

Considering first the one-year Treasury bill rate, the response decays quickly and becomes statistically non-significant after approximately one year, with broadly similar dynamics across both regimes. Turning to industrial production, the producer price index, and commodity prices, we find a sharp contrast across regimes. In the high-inventory regime, the response of industrial production is relatively small, with a peak fall of approximately 0.5%, while the price responses are essentially zero and statistically insignificant throughout. In the low-inventory regime, responses are considerably more pronounced: industrial production declines by more than 1% at peak, while prices fall by approximately 1%. As shown in Panel (b) of Figure 2, these differences are statistically significant at the 90% level. Moreover, both the excess bond premium and the inventory-to-sales ratio confirm their countercyclical nature, displaying a positive response in both regimes, with these responses again amplified in the low-inventory regime.

Results reveal a sharp asymmetry in monetary policy transmission. A tightening that occurs when firms hold high inventories appears nearly ineffective: responses in economic activity and prices are either small or statistically insignificant. By contrast, when firms enter the contractionary episode with low inventories, the same shock generates a pronounced and statistically significant contraction in industrial production and prices.



(a) State-dependent GIRFs to a monetary policy shock



(b) Difference in state-dependent GIRFs

Figure 2: Baseline specification

Note: Panel (a) shows the state-dependent GIRFs; Panel (b) shows the difference between the low- and the high-inventory regime responses. Each panel reports the response of the endogenous variables to a monetary policy shock, normalized so that the impact effect on the Treasury bill equals 25bp. Response of the Treasury bill and EBP are in basis point, while the response of the remain variables are in percent. Solid lines are posterior medians; dark and light shaded areas correspond to 68% and 90% posterior credible bands, respectively; blue (red) lines and bands correspond to the low- (high-) inventory regime. The state-dependent impact effect γ_i of the shock is identified using the proxy of [Bauer and Swanson \(2023\)](#).

2.1.3 Robustness checks

The baseline specification presented in the previous section involves several choices that, while defensible on econometric grounds, are ultimately arbitrary. To assess whether our findings are an artifact of these choices, we subject the baseline to a battery of perturbations. In each case, we modify one dimension of the specification at a time and verify that the asymmetric amplification result survives. For brevity, all results are relegated to Online Appendix E; here we describe the perturbations considered and summarize the main findings.

Monetary policy shocks Our baseline proxy for the monetary policy shock series follows [Bauer and Swanson \(2023\)](#). While this proxy represents one of the most rigorous recent advances, the literature offers several complementary approaches. We therefore re-estimate the model using three alternative instruments. First, we consider the series of [Miranda-Agrippino and Ricco \(2021\)](#) as extended by [Degasperi and Ricco \(2024\)](#), which removes the information component from monetary policy announcements, isolating a purer policy surprise. Second, we use the proxy of [Romer and Romer \(2004\)](#) as updated by [Wieland \(2021\)](#), identified via a narrative approach. Third, we use the series of [Aruoba and Drechsel \(2024\)](#), which extracts monetary policy shocks from Federal Reserve communications via a large language model.

See Figures E2, E3, and E4 of the Online Appendix for the results. The asymmetric amplification result is qualitatively robust across all three alternatives: both industrial production and prices display the same direction of asymmetry as in the baseline, with the low-inventory regime consistently showing a larger contraction than the high-inventory regime. Statistically, the response of industrial production remains strongly significant, similar to the baseline, while the price responses are somewhat less precisely estimated. Quantitatively, the magnitude of the effects is somewhat smaller than in the baseline for both variables. All in all, these findings reinforce the conclusion that the low-inventory

state amplifies monetary policy transmission, with the asymmetry in real activity being the most robustly estimated dimension.

Types of inventories The baseline inventory-to-sales ratio uses finished-goods inventories as the numerator. While this is the most direct measure of goods available for sale, we assess robustness along three dimensions. First, we subtract backlogs of orders from the numerator, so that the state variable captures inventories of goods immediately ready for delivery rather than production already committed to future orders. Second, we consider two alternative inventory aggregates: input inventories (raw materials and work-in-progress in manufacturing) and total inventories (the sum of all types of inventories), which, while not the exact object of interest, provide broader or upstream measures of firms' inventory positions. Third, to address concerns that the regime classification may reflect contemporaneous demand conditions rather than firms' pre-existing inventory positions, we replace the last-quarter sales denominator with the last two quarters and over the last year, respectively.

See Online Appendix Figures E5, E6, E7, E8, and E9 for the estimated differences in state-dependent GIRFs under each alternative. The asymmetric amplification result holds across all the cases: contractionary monetary policy consistently generates statistically significant larger contractions in the low-inventory regime. Effects on prices are in some cases somewhat less pronounced than in the baseline.

Threshold delay and classification rule Two aspects of the regime-classification scheme are assessed. First, the delay d entering the threshold condition $y_{n,t-d} \leq y_n^*$ is selected endogenously by the discrete Gibbs step of the MCMC, with the baseline posterior concentrating on $d = 2$. We verify that the asymmetric amplification result is not an artifact of this particular value by fixing $d = 1$ and $d = 3$ in turn. Second, in the baseline y_n^* is treated as an unknown parameter and estimated jointly with the rest of the model via the Metropolis–Hastings step, with a prior centered at zero. We additionally consider a non-Bayesian classification rule that fixes $y_n^* = 0$, which equals the unconditional mean of the

band-pass-filtered IS ratio by construction, classifying regimes deterministically according to whether the state variable lies above or below its long-run trend. The asymmetric amplification result survives in all cases.

Sign and size of the shock A key advantage of generalized impulse response functions, relative to local projections, is the ability to condition on the size and sign of the shock. Local projections estimate average effects irrespective of these characteristics [Gonçalves et al., 2024](#), whereas the TVAR framework allows responses to be evaluated at any prescribed shock magnitude and direction [Francis et al., 2023](#). The baseline normalizes the shock to a 25 basis point on the policy rate. We assess robustness by considering a larger contractionary shock of 100 basis points and an expansionary shock of -25 basis points.

Figures [E10](#) and [E11](#) of the Online Appendix present the implied differences in GIRFs across regimes under these two alternatives. The asymmetric amplification is highly robust and carries an important implication for the local projection evidence presented in Section [2.1.4](#): since the state-dependent asymmetry is approximately invariant to the shock's size and sign, the average effects captured by local projections constitute a reliable approximation to the underlying regime-specific dynamics.

Endogenous variables The baseline VAR includes the producer price index (PPI) as a nominal variable. To assess whether the results extend to alternative price and quantity measures, we consider two extensions. First, we replace the PPI with the consumer price index (CPI), which more directly captures the cost of living for a representative household. Second, we add the unemployment rate as an additional endogenous variable to verify that the responses of industrial production are mirrored in the labor market.

See Figures [E12](#) and [E13](#) of the Online Appendix for the results. The CPI response is qualitatively consistent with the PPI, confirming the asymmetric amplification pattern, though less pronounced. The unemployment rate shows a difference in response between the two regimes of almost 15 basis points, consistent with the industrial production dynamics.

2.1.4 Local Projections

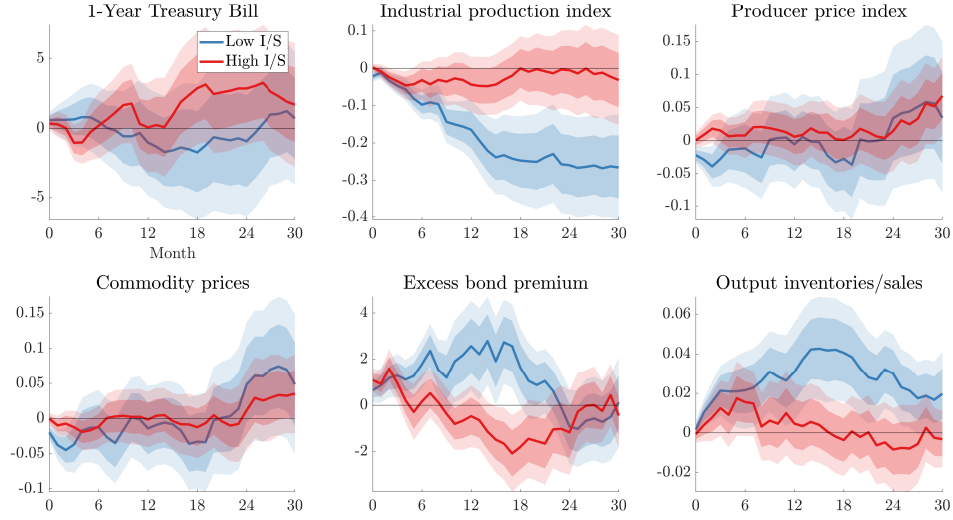
A natural question is whether the results are specific to the threshold VAR or continue to hold under alternative estimators. VAR models impose a particular structure on the economy, and if the data-generating process deviates from this structure, inference may be distorted. Local projections (Jordà, 2005) provide a complementary approach by directly projecting outcomes on the shock of interest, thereby relaxing some of these parametric restrictions. Despite greater sampling uncertainty (Plagborg-Møller and Wolf, 2021) and potential bias in state-dependent settings (Gonçalves et al., 2024), local projections remain a useful robustness check: if they deliver qualitatively similar results, this confirms that the findings are not an artifact of the VAR's assumptions.

For each endogenous variable $y_{i,t}$ and horizon h , we estimate:

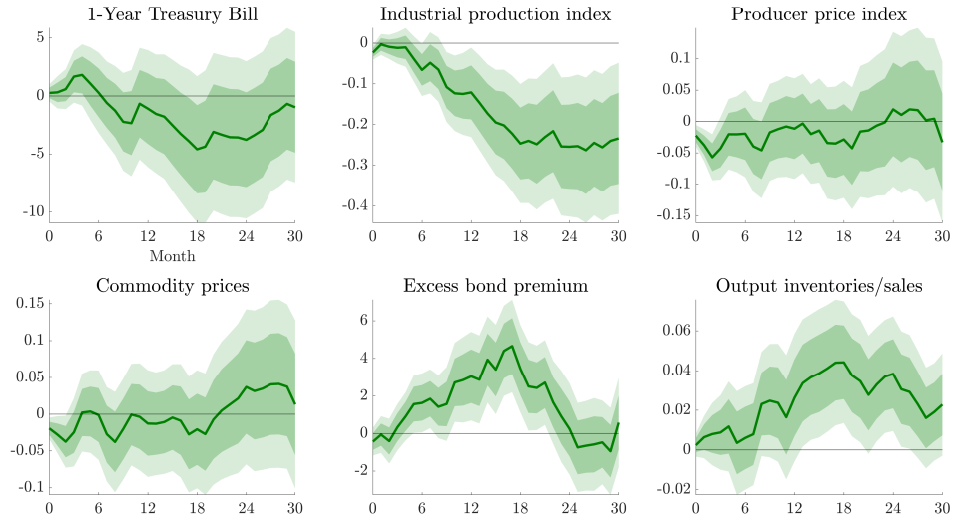
$$y_{i,t+h} - y_{i,t-1} = \theta_{i,h}^L S_t \hat{\varepsilon}_t + \theta_{i,h}^H (1 - S_t) \hat{\varepsilon}_t + \Gamma_h X_t + u_{t,h}, \quad (4)$$

where $y_{i,t+h} - y_{i,t-1}$ is the cumulative change from $t - 1$ to $t + h$ in variable i and $\hat{\varepsilon}_t$ is the monetary policy shock proxy. The binary variable S_t is the regime indicator, equal to one if the underlying inventory-to-sales ratio is below its mean and zero otherwise. This ratio is defined consistently with the baseline VAR as the filtered finished-goods inventory-to-sales ratio, delayed by one quarter. The coefficients $\theta_{i,h}^L$ and $\theta_{i,h}^H$ capture the state-dependent impulse responses of variable i at horizon h in the low- and high-inventory regimes, respectively. The control vector X_t includes a constant, 12 lags of the shock and of the first differences of all endogenous variables of the baseline VAR, together with their interactions with S_t . Standard errors are heteroskedasticity-robust (Montiel Olea and Plagborg-Møller, 2021).

Figure 3 presents the state-dependent impulse response functions of all variables in the baseline specification to a contractionary monetary policy shock, Panel (a), together with



(a) State-dependent IRFs to a monetary policy shock



(b) Difference in state-dependent IRFs

Figure 3: Local projections

Note: Panel (a) shows the state-dependent IRFs; Panel (b) shows the difference between the low- and the high-inventory regime responses. Each panel reports the response of the endogenous variables to a standard-deviation monetary policy shock. Response of the Treasury bill and EBP are in basis point, while the response of the remain variables are in percent. Solid lines are point estimates; dark and light shaded areas correspond to 68% and 90% confidence bands, respectively; blue (red) lines and bands correspond to the low- (high-) inventory regime. The effects of the shock are identified using the proxy of [Bauer and Swanson \(2023\)](#).

the implied differences in responses across regimes, Panel (b).⁴ Solid lines are point es-

⁴ We normalize the shock to one standard deviation rather than to a 25 basis point impact on the one-year Treasury bill because the [Bauer and Swanson \(2023\)](#) proxy has a remarkably smaller and noisier effect

timates; dark and light shaded areas correspond to 68 and 90 percent confidence bands, respectively; blue and red lines and bands correspond to the low- and high-inventory regimes. We display responses up to 30 months: given the shorter sample available for local projections, the estimated responses tend to become less meaningful at longer horizons.

The qualitative picture is remarkably consistent across variables. For industrial production, the shock has a relatively small effect in the high-inventory regime but produces a much larger decline in the low-inventory regime, and this difference is highly significant. For the producer price index and commodity prices, responses are not statistically significant in the high-inventory regime but turn negative and significant in the low state, with a significant difference over the first quarter. Similarly, the excess bond premium and the inventories-to-sales ratio deliver responses closely in line with the baseline specification, again with a statistically significant asymmetry across regimes at the 90 percent level.

Overall, the fact that a different sample period and a different estimator continue to deliver qualitatively consistent asymmetric amplification provides strong confirmation of the baseline results. These exercises also serve as a natural bridge to the next section, where we show that the state-dependent amplification is not limited to time variation alone but survives when identification exploits cross-sectional variation across sectors.

2.2 Sector-level evidence

In this section, we explore whether the state-dependent effect of monetary policy extends beyond the aggregate dynamics documented above. There are two reasons why we do so.

on this variable in the local projection setting. Normalizing to a 25 basis point impact would therefore produce implausibly large responses in the remaining variables, whereas a one-standard-deviation shock yields responses that are quantitatively in line with the baseline VAR. The proxy appears weaker in the local projection than in the VAR for a structural reason. In the VAR, the reduced-form system is estimated over the full sample for which the endogenous variables are available, and the proxy enters only in the structural step that identifies the contemporaneous impact of the shock. The reduced-form innovations are however estimated over a longer sample and are allegedly more precise, even though the proxy itself is observed only over a shorter period. In the local projection, by contrast, the instrument enters the regression directly at each horizon, so the entire estimation is constrained to the shorter sample over which the instrument is available. Fewer observations mechanically attenuate the relevance of the proxy.

First, the aggregate inventory-to-sales ratio is inherently counter-cyclical, raising the concern that the state-dependent effect implicitly reflects the sensitivity of monetary policy to the business cycle. Sector-level evidence may help assuage this concern: by controlling for the overall state of the economy through time fixed effects, it isolates variation in inventory positions that is idiosyncratic to each sector and therefore orthogonal to aggregate cyclical conditions. Second, we want to better understand the mechanism through which inventories shape the transmission of monetary policy. Specifically, we ask whether the state-dependent amplification requires a general equilibrium mechanism to operate or whether it can also arise independently of such effects. The answer informs the type of theory needed to rationalize the empirical results.

The underlying dataset is an unbalanced panel of 35 three-digit NAICS industries observed at monthly frequency. We draw on three data sources: the U.S. Census Bureau’s industry surveys to measure inventories and sales; the Federal Reserve’s G.17 statistical release to measure industrial production; and the Bureau of Labor Statistics’ Producer Price Index program to measure prices. See Appendix A.2 for further details on data construction and sample coverage.

For each sector j , endogenous variable i , and horizon h , we estimate the state-dependent local projection

$$y_{i,j,t+h} - y_{i,j,t-1} = \delta_h \hat{\varepsilon}_t S_{j,t} + \Gamma_h X_{j,t} + \gamma_j + \mu_t + u_{i,j,t+h}, \quad (5)$$

where $y_{i,j,t+h} - y_{i,j,t-1}$ is the cumulative log change in variable i for sector j from month $t - 1$ to month $t + h$; $\hat{\varepsilon}_t$ is the [Bauer and Swanson \(2023\)](#) monetary policy shock; and the binary variable $S_{j,t}$ is the sector-specific regime indicator, equal to one if the underlying inventory-to-sales ratio is below its mean and zero otherwise. This ratio is defined as the band-pass-filtered sector- j finished-goods inventory-to-sales ratio, delayed by one quarter. The coefficient δ_h captures the differential dynamic response of sector-level variable i at

horizon h between low- and high-inventory states following a contractionary shock. The intercept γ_j is a sector fixed effect and μ_t is a time fixed effect. Because $\hat{\varepsilon}_t$ varies only over time, it is collinear with μ_t and its aggregate level effect is absorbed; identification of δ_h therefore rests exclusively on the cross-sectional variation in $S_{j,t}$, comparing sectors that face the same monetary policy shock at the same date but differ in their inventory position. The control vector $X_{j,t}$ includes $p = 12$ uninteracted lags of the first-differenced sector outcome, together with p lags of the shock, the first-differenced sector outcome, and aggregate industrial production growth, all interacted with $S_{j,t}$. Standard errors are clustered at the monthly level.

Importantly, sector and time fixed effects ensure that δ_h is identified from differences in the response between two sector-time observations whenever those observations differ in their value of $S_{j,t}$. This includes two main sources of variation: the same sector switching inventory state at a different point in time or two different sectors in different states at the same point in time.⁵ Thus, sector and time fixed effects controls for two main confounding channels. The sector fixed effect γ_j ensures that δ_h does not capture time-invariant structural differences across industries: sectors that structurally carry fewer inventories may also be more or less sensitive to monetary policy shocks for reasons unrelated to their current inventory position, and the sector fixed effect completely excludes this option. The time fixed effect μ_t ensures that δ_h does not capture sector-invariant aggregate variation over time that is correlated with the inventory-to-sales ratio, thereby assuaging the concern that the results obtained at the aggregate level are a byproduct of the state of the economy rather than the level of inventories.

Figure 4 plots $\hat{\delta}_h$ from equation (5), the estimated difference in impulse responses between the low- and the high-inventory state. Solid lines are point estimates; dark and light shaded areas correspond to 68% and 90% confidence bands, respectively. The qualitative pattern closely mirrors the aggregate results: a contractionary monetary policy shock

⁵ More generally, both the sector and the time period can differ simultaneously, yielding a difference-in-differences comparison.

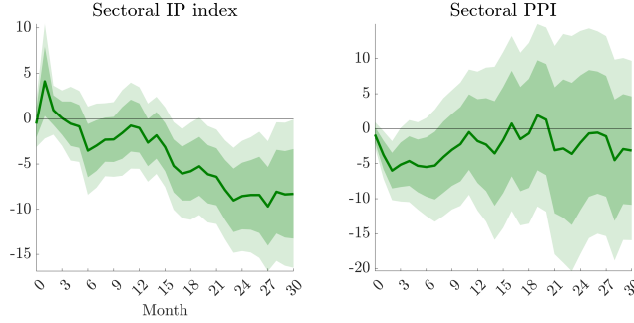


Figure 4: Sectoral evidence – Difference in state-dependent IRFs

Note: The figure plots $\hat{\delta}_h$ from equation (5), the differential impulse response between the low- and the high-inventory state, estimated on the panel of 35 three-digit NAICS industries with sector and time fixed effects. The left panel reports results for the sectoral IP index; the right panel for the sectoral PPI. The monetary policy shock is from [Bauer and Swanson \(2023\)](#). Solid lines are point estimates; dark and light shaded areas correspond to 68% and 90% confidence bands. A negative value means that low-inventory sectors have a more negative response than high-inventory sectors in response to a contractionary shock.

generates a more negative response of both industrial production and producer prices in low-inventory sectors than in high-inventory sectors.

2.3 Firm-level evidence

The aggregate and sectoral exercises both define the inventory state from the inventory-to-sales ratio measured at a relatively coarse level of aggregation. This raises a residual endogeneity concern: the inventory-to-sales ratio is itself an equilibrium object, so periods or sectors with low inventories may differ along other dimensions that independently generate state dependence in monetary transmission. If so, the asymmetry documented above could reflect those confounding forces rather than the inventory position per se. The firm-level analysis is designed to address this concern directly. By comparing firms within the same industry and the same quarter, we can absorb *all* aggregate and industry-wide time-varying forces—the business cycle, credit conditions, sectoral demand and supply shocks, and any general equilibrium response to monetary policy—and isolate the role of a firm’s own inventory position. The exercise also follows naturally from a large literature that studies the investment channel of monetary policy using firm-level data and

high-frequency shocks, most closely [Jeenas \(2024\)](#), [Ottonello and Winberry \(2020\)](#), and [Cloyne et al. \(2023\)](#).

Data and inventory state. We use quarterly Compustat data on U.S. non-financial firms over 1988–2023, the sample for which the [Bauer and Swanson \(2023\)](#) monetary policy shock is available. Following [Jeenas \(2024\)](#) and [Cloyne et al. \(2023\)](#), we drop firm-quarters with acquisitions exceeding five percent of total assets, trim the key variables at the first and ninety-ninth percentiles within each year, and keep firms with at least twenty quarters of observations, leaving roughly 6,000 firms. We measure a firm’s inventory position by its total inventory-to-sales ratio and classify firm j as being in the *low-inventory state* in quarter t whenever this ratio lies below the median ratio of its three-digit industry in that quarter:

$$\mathbb{1}_{j,t}^{\text{low}} = \mathbb{1} \left\{ \left(\frac{\text{inventories}}{\text{sales}} \right)_{j,t} < \text{med}_{k \in s(j)} \left(\frac{\text{inventories}}{\text{sales}} \right)_{k,t} \right\}, \quad (6)$$

where $s(j)$ denotes the three-digit SIC industry of firm j . Defining the state relative to the contemporaneous industry median has two advantages. First, it requires no time-series filtering of the firm’s inventory series, providing a robustness check on the filtering choices made in the aggregate analysis. Second, it nets out permanent and slow-moving industry-level differences in inventory practices, so that the state captures whether a firm is temporarily lean relative to its peers. The results are unchanged if the state is instead defined from a within-firm Hamilton filter, or if the discrete indicator is replaced by the continuous distance of the firm’s ratio from its industry median.⁶

Specification. For each outcome and horizon $h = 0, 1, \dots, 20$ quarters we estimate the state-dependent local projection

$$k_{j,t+h} - k_{j,t-1} = \beta_h \hat{\varepsilon}_t \mathbb{1}_{j,t-1}^{\text{low}} + \mathbf{\Gamma}'_h \mathbf{X}_{j,t-1} + \alpha_{j,h} + \alpha_{s(j),t,h} + u_{j,t+h}, \quad (7)$$

⁶ These alternatives are reported in the Online Appendix.

where $k_{j,t}$ is the log of firm j 's tangible capital, so the left-hand side is the cumulative change in (log) capital between $t - 1$ and $t + h$; $\hat{\varepsilon}_t$ is the [Bauer and Swanson \(2023\)](#) monetary policy shock; and $\mathbb{1}_{j,t-1}^{\text{low}}$ is the lagged low-inventory indicator from (6). The term $\alpha_{j,h}$ is a firm fixed effect and $\alpha_{s(j),t,h}$ is an industry $\times$ quarter fixed effect. The latter is central to the design: because the monetary policy shock is common across firms in a given quarter, the industry $\times$ quarter fixed effect absorbs its average (level) effect, so that β_h is identified purely from the comparison of low- versus high-inventory firms operating in the same industry at the same date. A negative β_h therefore means that, faced with an identical contractionary shock, firms that are temporarily lean cut capital by more than their industry peers. Standard errors are two-way clustered by firm and by quarter.

The control vector $\mathbf{X}_{j,t-1}$ follows the convention in this literature of interacting the state with the dynamic controls. In our baseline specification (Panel (a) of [Figure 5](#)) it contains the lagged state indicator, four lags of the aggregate variables of the time-series analysis (the growth rates of industrial production, the producer price index, and commodity prices, the one-year Treasury rate, the shock, and the excess bond premium) and four lags of the firm's own log-capital and log-sales growth, together with the interactions of all of these terms with the lagged state indicator.⁷

In our preferred specification (Panel (b)) we additionally control for a rich set of firm characteristics and, crucially, for their interactions with the monetary policy shock. Specifically, we add one lag of (log) total assets, Tobin's Q, leverage, age, the quick ratio, the liquidity ratio, dividends, and the distance to default, and we interact the shock $\hat{\varepsilon}_t$ with each of these lagged characteristics. These interactions are exactly the channels emphasized by [Ottonello and Winberry \(2020\)](#), [Jeenas \(2024\)](#), and [Gertler and Gilchrist \(1994\)](#)—leverage, default risk, balance-sheet liquidity, and firm size. By allowing the firm-level response to the shock to vary freely with each of them, we ensure that β_h captures the role

⁷ The uninteracted aggregate terms are collinear with the industry $\times$ quarter fixed effects and are absorbed; they are retained in the regression so that the state interactions are properly orthogonalized.

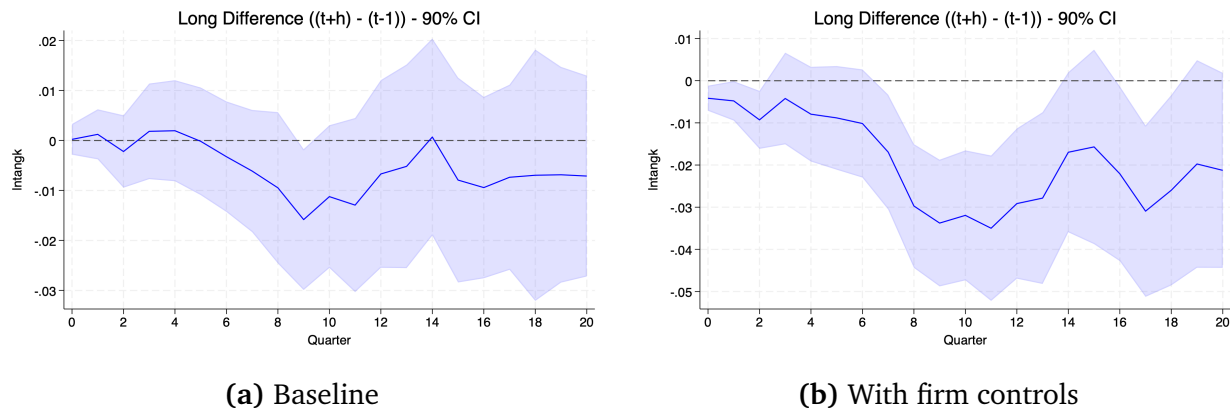


Figure 5: Firm-level state-dependent response of tangible capital

Note: Each panel plots the coefficient β_h from the local projection (7), i.e. the differential cumulative response of (log) tangible capital of low-inventory firms relative to high-inventory firms in the same three-digit industry and quarter, following a contractionary monetary policy shock identified by the proxy of [Bauer and Swanson \(2023\)](#). The horizon is in quarters and the response is the long difference $k_{j,t+h} - k_{j,t-1}$. Panel (a) is the baseline specification; Panel (b) adds lagged firm characteristics and their interactions with the shock, following [Ottonello and Winberry \(2020\)](#) and [Jeenas \(2024\)](#). All specifications include firm and industry $\times$ quarter fixed effects, with standard errors two-way clustered by firm and quarter. Solid lines are point estimates and shaded areas are 90% confidence bands.

of the inventory position and not other dimensions of firm heterogeneity already known to shape the transmission of monetary policy.

Figure 5 reports the estimated differential response β_h . In both specifications the response of tangible capital is negative throughout, building to roughly -3 to -3.5 percent around two to three years after the shock and statistically significant at the 90% level over the medium term. Low-inventory firms therefore contract investment substantially more than otherwise comparable firms in the same industry and quarter. Reassuringly, the estimate is essentially unchanged when we move from the baseline in Panel (a) to the specification in Panel (b) that controls for the full battery of firm characteristics and shock interactions: the inventory-state amplification is not absorbed by leverage, liquidity, size, or default risk. Because the industry $\times$ quarter fixed effects strip out every aggregate and sectoral time-varying force, this evidence isolates the firm's own inventory position as the operative margin and corroborates, at the most disaggregated level, the asymmetric amplification documented in the aggregate and sectoral analyses.

Finally, Table 1 reports, for the firm characteristics that enter the regression, summary statistics by inventory state. The two states are not balanced: relative to high-inventory firms, low-inventory firms are on average larger (higher sales and total assets), more liquid (higher quick ratio), safer (a greater distance to default), carry a higher Tobin’s Q, and pay more dividends, while their leverage is essentially identical. Three considerations address the econometric concern this imbalance might raise. First, the differences reported in the table are unconditional, whereas the specification includes firm fixed effects: the characteristics that differ most across states—size, age, and default risk—are slow-moving firm traits that are largely absorbed by $\alpha_{j,h}$, so that identification comes from a given firm being temporarily lean relative to its industry rather than from comparing structurally different firms. Second, the preferred specification additionally controls for these characteristics and for their interactions with the shock, letting the firm-level response vary freely along each dimension the investment-channel literature emphasizes; the estimated amplification is essentially unchanged, as the comparison of Panels (a) and (b) of Figure 5 makes clear. Third, the selection runs against the result: Gertler and Gilchrist (1994), Ottonello and Winberry (2020), and Jeenas (2024) predict that larger, more liquid, less leveraged, safer, and dividend-paying firms are *less* responsive to monetary policy, yet these are precisely the firms that populate the low-inventory state, which nonetheless displays the stronger response. If anything, the imbalance biases the estimated amplification toward zero.

3 Theory

We embed inventories into an otherwise standard New Keynesian model following Lubik and Teo (2012). The economy consists of a continuum of monopolistically competitive firms, a representative household, and a monetary authority that implements monetary policy. The central departure from the textbook framework is that firms hold a stock of finished goods which simultaneously facilitates sales and generates a state-dependent cost of price adjustment.

Table 1: Summary statistics by inventory state

	Low inventory ($\mathbb{1}^{\text{low}} = 1$)	High inventory ($\mathbb{1}^{\text{low}} = 0$)	Difference (low – high)
Inventory-to-sales	0.090	0.216	−0.126***
Log sales	4.536	3.959	0.576***
Log total assets	5.778	5.350	0.428***
Tobin’s Q	1.766	1.499	0.266***
Leverage	0.256	0.260	−0.005***
Quick ratio	0.979	0.842	0.138***
Distance to default	6.548	5.460	1.087***
Age	16.419	15.817	0.603***
Dividends	13.845	7.249	6.597***

Note: Unconditional means by inventory state, restricted to the variables that enter the local projection (7). The low-inventory state is defined in (6) as a firm’s total inventory-to-sales ratio falling below its three-digit SIC industry median in the quarter. The last column reports the difference in means (low minus high), tested with a two-sample t -test; *** denotes significance at the 1% level. The number of non-missing pooled firm-quarters ranges from about 220,000 (distance to default) to 312,000 (inventory-to-sales, sales, and total assets).

3.1 Firms

There is a unit continuum of firms, indexed by $i \in [0, 1]$. Firm i produces output $y_t(i)$ using labor $h_t(i)$ according to the linear production function

$$y_t(i) = z_t h_t(i), \quad (8)$$

where z_t is aggregate total factor productivity, which follows the exogenous AR(1) process $\ln z_t = \rho_z \ln z_{t-1} + \varepsilon_{z,t}$ with $|\rho_z| < 1$.

Stock-elastic demand. Following [Bils and Kahn \(2000\)](#) and [Lubik and Teo \(2012\)](#), we assume that a firm’s stock of available goods facilitates sales. Households aggregate the differentiated goods of all firms using a modified Dixit-Stiglitz aggregator that allows inventory holdings to shift demand:

$$s_t = \left[\int_0^1 \left(\frac{a_t(i)}{a_t} \right)^\mu s_t(i)^{(\theta-1)/\theta} di \right]^{\theta/(\theta-1)}, \quad (9)$$

where s_t is aggregate sales, $s_t(i)$ is firm i 's sales, $a_t(i)$ is the stock of goods available for sale at firm i , and $a_t = \int_0^1 a_t(i) di$ is the aggregate stock. The parameter $\theta > 1$ is the elasticity of substitution across differentiated goods, and $\mu > 0$ is the elasticity of demand with respect to the firm's relative inventory stock. A higher stock of goods facilitates sales: households are more likely to find and purchase a variety when it is abundantly available. The parameter μ therefore captures the *stock-elastic* nature of demand.

Cost minimization by households implies the demand function for firm i 's goods:

$$s_t(i) = \left(\frac{a_t(i)}{a_t} \right)^\mu \left(\frac{P_t(i)}{P_t} \right)^{-\theta} s_t, \quad (10)$$

where $P_t(i)$ is the price set by firm i and P_t is the aggregate price index:

$$P_t = \left[\int_0^1 \left(\frac{a_t(i)}{a_t} \right)^\mu P_t(i)^{1-\theta} di \right]^{1/(1-\theta)}. \quad (11)$$

Equation (10) shows that sales are increasing in the firm's relative inventory holding and decreasing in its relative price. In a symmetric equilibrium all firms hold the same stock, so $a_t(i)/a_t = 1$ and the relative stock term drops out: inventory stocks raise sales only insofar as a firm's holdings exceed the aggregate level. What remains, however, is a crucial general equilibrium implication: the level of the aggregate stock a_t enters the model's dynamics through the firm's optimality conditions.

Inventory law of motion. The stock of goods available for sale evolves according to:

$$a_t(i) = y_t(i) + (1 - \delta)(a_{t-1}(i) - s_{t-1}(i)), \quad (12)$$

where $\delta \in (0, 1)$ is the rate at which unsold inventory depreciates or becomes obsolete between periods. Current production $y_t(i)$ is added to the inherited stock of unsold goods from last period, $(1 - \delta)(a_{t-1}(i) - s_{t-1}(i))$, to form the stock available for today's sales. Equation (12) decouples production from contemporaneous sales: firms can produce more

than they sell in one period and carry the excess into the next, and they can sell more than they produce by drawing down previously accumulated stocks.

Price adjustment costs. Firm i faces a quadratic cost of adjusting its nominal price:

$$\frac{\phi}{2} \left(\frac{P_t(i)}{\pi P_{t-1}(i)} - 1 \right)^2 s_t, \quad (13)$$

where $\phi > 0$ governs the magnitude of the adjustment cost, and $\pi \geq 1$ is the steady-state gross inflation rate. Critically, the adjustment cost is measured in units of *aggregate sales* s_t rather than output. This specification reflects the idea that the deadweight cost of a suboptimal price is proportional to the volume of transactions affected by that price: a firm selling many units at a wrong price suffers a larger loss than one selling few units. As we discuss below, this feature is at the heart of the model's state-dependent transmission mechanism.

Profit maximization. Firm i chooses its price $P_t(i)$, labor input $h_t(i)$, and stock $a_t(i)$ to maximize the expected discounted stream of real profits:

$$E_t \sum_{\tau=0}^{\infty} \rho_{t,t+\tau} \left[\frac{P_{t+\tau}(i) s_{t+\tau}(i)}{P_{t+\tau}} - \frac{W_{t+\tau} h_{t+\tau}(i)}{P_{t+\tau}} - \frac{\phi}{2} \left(\frac{P_{t+\tau}(i)}{\pi P_{t+\tau-1}(i)} - 1 \right)^2 s_{t+\tau} \right], \quad (14)$$

subject to the production function (8), the demand function (10), and the inventory law of motion (12). Here W_t is the nominal wage and $\rho_{t,t+\tau}$ is the stochastic discount factor used by the firm to evaluate future profit streams.

First-order conditions. The optimality conditions associated with the firm's three choice variables yield three key equilibrium relationships. The first-order condition for the price $P_t(i)$ gives a modified New Keynesian Phillips curve:

$$\phi \left(\frac{P_t(i)}{\pi P_{t-1}(i)} - 1 \right) \frac{s_t}{\pi P_{t-1}(i)} = (1 - \theta) \frac{s_t(i)}{P_t}$$

$$+ E_t \rho_{t,t+1} \left[\phi \left(\frac{P_{t+1}(i)}{\pi P_t(i)} - 1 \right) \frac{s_{t+1} P_{t+1}(i)}{\pi P_t(i)^2} + (1 - \delta) \theta \frac{s_t(i)}{P_t(i)} m_{c_{t+1}}(i) \right], \quad (15)$$

where $m_{c_t}(i)$ is the Lagrange multiplier on the demand constraint (10), interpretable as real marginal cost. Note that the current and forward-looking adjustment-cost terms scale with *aggregate* sales, s_t and s_{t+1} , since the Rotemberg cost (13) is measured in units of aggregate sales and the firm takes these as given; it is the firm's own sales $s_t(i)$ that enter only the markup and marginal-cost terms. Beyond this, equation (15) resembles the standard Rotemberg pricing condition but with a forward-looking term involving future marginal cost, weighted by the carryover factor $(1 - \delta)$. This additional term arises because today's pricing decision determines the current inventory, which affects future sales and hence future production costs.

The first-order condition for labor input equates the real wage to the marginal revenue product of labor:

$$\frac{W_t}{P_t} = z_t m_{c_t}(i). \quad (16)$$

The first-order condition for the inventory stock $a_t(i)$ is an intertemporal Euler equation for inventory investment:

$$m_{c_t}(i) = \mu \frac{P_t(i)}{P_t} \frac{s_t(i)}{a_t(i)} + (1 - \delta) \left(1 - \mu \frac{s_t(i)}{a_t(i)} \right) E_t \rho_{t,t+1} m_{c_{t+1}}(i). \quad (17)$$

Equation (17) is central to the model's dynamics. It states that the marginal cost of producing one unit today equals the sum of two benefits: (i) the direct revenue gain from immediately selling that unit, proportional to the sales-to-stock ratio $\gamma_t(i) \equiv s_t(i)/a_t(i)$ weighted by μ ; and (ii) the discounted future cost saved by having more stock available next period, which reduces the need to produce tomorrow. When $\mu = 0$, the stock-elastic demand feature is shut off and marginal cost reduces to the standard present value of future production costs. As μ rises, current sales-intensity γ_t gains weight in the deter-

mination of current marginal cost, making the latter more sensitive to the firm's current inventory position.

3.2 Households

There is a representative household with expected lifetime utility:

$$E_0 \sum_{t=0}^{\infty} \beta^t \zeta_t \left[\ln c_t - \frac{h_t^{1+\eta}}{1+\eta} \right], \quad (18)$$

where c_t is consumption, h_t is aggregate labor hours, $\beta \in (0, 1)$ is the discount factor, $\eta \geq 0$ is the inverse Frisch elasticity of labor supply, and ζ_t is a preference shock following $\ln \zeta_t = \rho_\zeta \ln \zeta_{t-1} + \varepsilon_{\zeta,t}$. The household supplies labor to firms at the nominal wage W_t , earns dividend income D_t from firm ownership, and holds one-period discount bonds B_t at a price $1/R_t$, where R_t is the gross nominal interest rate. Its flow budget constraint is:

$$P_t c_t + \frac{B_t}{R_t} \leq B_{t-1} + W_t h_t + D_t. \quad (19)$$

The household's first-order conditions are the intratemporal labor supply condition:

$$h_t^\eta = \frac{\zeta_t W_t}{c_t P_t}, \quad (20)$$

and the intertemporal consumption Euler equation:

$$\frac{\zeta_t}{c_t} = \beta R_t E_t \left[\frac{\zeta_{t+1} P_t}{c_{t+1} P_{t+1}} \right]. \quad (21)$$

Equation (20) equates the real wage to the marginal rate of substitution between leisure and consumption. Equation (21) is the standard bond-pricing condition linking the nominal interest rate to expected consumption growth and inflation.

3.3 Monetary Authority and Market Clearing

The monetary authority sets the nominal interest rate according to a Taylor-type feedback rule:

$$\tilde{R}_t = \rho_R \tilde{R}_{t-1} + \psi_1 \tilde{\pi}_t + \psi_2 \tilde{y}_t + \varepsilon_{R,t}, \quad (22)$$

where tildes denote log-deviations from steady state, $\rho_R \in [0, 1)$ is an interest smoothing parameter, $\psi_1 > 1$ and $\psi_2 \geq 0$ are the responses to inflation and output, and $\varepsilon_{R,t}$ is a monetary policy shock.

We impose a symmetric equilibrium in which all firms make identical choices, so that firm-specific indices can be dropped. Bonds are in zero net supply, $B_t = 0$. Goods market clearing requires that aggregate sales equal household consumption plus the real resource cost of price adjustment:

$$s_t = c_t + \frac{\phi}{2} \left(\frac{\pi_t}{\pi} - 1 \right)^2 c_t, \quad (23)$$

where $\pi_t = P_t/P_{t-1}$ is gross inflation. The labor market clears when aggregate hours equal aggregate output divided by productivity: $h_t = y_t/z_t$.

3.4 Generalized Impulse Response Functions

To study state-dependent dynamics we compute Generalized Impulse Response Functions (GIRFs), simulating the nonlinear model from two initial conditions — one with a high inventory-to-sales ratio and one with a low ratio — and measuring the difference in variable paths with and without a contractionary monetary policy shock, normalized to a 25-basis-point increase in the nominal interest rate at impact.

Figure 6 reports the GIRFs for the baseline calibration. We set $\beta = 0.99$, $\theta = 11$ (10% steady-state markup), $\eta = 1$, ϕ calibrated to imply an inflation coefficient in the linearized Phillips curve of 0.1 (as in Ireland 2004), $\mu = 0.37$ and $\delta = 0.01$ following Lubik and Teo (2012), and Taylor rule parameters $\psi_1 = 1.5$, $\psi_2 = 0$, $\rho_R = 0.7$.

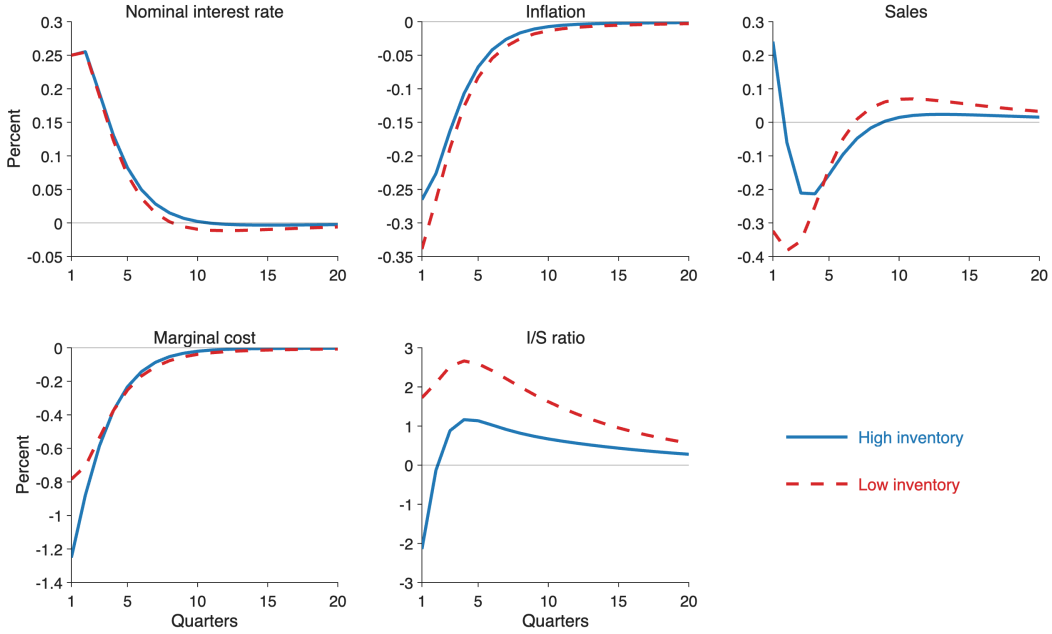


Figure 6: Generalized impulse responses to a contractionary monetary policy shock (normalized to 25 basis points at impact) in the Lubik-Teo model. Blue solid: high initial inventory-to-sales ratio. Red dashed: low initial inventory-to-sales ratio. Horizons in quarters.

The figure reveals a clear asymmetry across inventory states. *Inflation* falls more sharply in the low-inventory state (red dashed) than in the high-inventory state (blue solid). *Sales* decline more in the low-inventory state as well. The *marginal cost* response is, by contrast, larger in the high-inventory state. The *inventory stock* rises in both states as firms increase production following the shock, but the build-up is larger in the low-inventory state. We now develop the mispricing channel, which explains these patterns.

3.5 The Mispricing Channel

The GIRF asymmetries are driven by what we call the *mispricing channel*. To expose the mechanism, consider the symmetric equilibrium in which $a_t(i)/a_t = 1$, $P_t(i)/P_t = 1$, and $s_t(i)/a_t(i) = \gamma_t \equiv s_t/a_t$. The inventory Euler equation (17) becomes:

$$mc_t = \mu \gamma_t + (1 - \delta)(1 - \mu \gamma_t) E_t \rho_{t,t+1} mc_{t+1}, \quad (24)$$

and the price-setting condition (15) simplifies to:

$$\phi \left(\frac{\pi_t}{\pi_{ss}} - 1 \right) \frac{\pi_t}{\pi_{ss}} = (1 - \theta) + E_t \rho_{t,t+1} \left[\phi \left(\frac{\pi_{t+1}}{\pi_{ss}} - 1 \right) \frac{s_{t+1}}{s_t} \frac{\pi_{t+1}}{\pi_{ss}} + (1 - \delta) \theta m c_{t+1} \right]. \quad (25)$$

Equation (25) differs from the standard Rotemberg Phillips curve in two ways: the forward-looking adjustment-cost term carries the ratio s_{t+1}/s_t , scaling future price-adjustment costs by current sales; and an additional term $(1 - \delta)\theta m c_{t+1}$ arises because today's pricing decision determines current inventory, which feeds into future production costs through (24).

The source of state dependence is the forward-looking adjustment-cost term in (25), which is scaled by the ratio of expected future to current aggregate sales, s_{t+1}/s_t . Because the Rotemberg cost (13) is measured in units of *aggregate* sales rather than the firm's own sales, the weight that firms place on future relative to current price adjustment rises whenever expected future sales are high relative to today's. The inventory-to-sales position governs precisely this ratio. When the economy carries a large stock relative to current sales, sales are expected to grow as the stock is drawn down, so $E_t[s_{t+1}/s_t]$ is high; when inventories are lean, expected future sales are low relative to current sales and $E_t[s_{t+1}/s_t]$ is small. The inventory state therefore determines how forward-looking the firm's pricing decision is, and hence how much of the adjustment to a shock is borne by the current price versus deferred to the future.

High inventory state. When the economy enters a contractionary episode with a high inventory-to-sales ratio a_t/s_t (low γ_t), expected future aggregate sales are high relative to current sales, so $E_t[s_{t+1}/s_t]$ is large and the forward-looking term in (25) receives substantial weight. The firm's pricing decision becomes more forward-looking: it defers adjustment to future periods, so the current price moves little and the inflation response to the tightening is muted. The Euler equation (24) adds a counteracting force on the cost side: with low γ_t , current marginal cost loads heavily on discounted future marginal cost, so the contractionary shock, by lowering expected future wages, gets strongly capitalized into

mc_t today. This marginal-cost channel operates in the opposite direction to the mispricing channel, and explains why the model generates a larger mc_t response in the high-inventory state despite the smaller inflation response there.

Low inventory state. When inventories are low (high γ_t), the logic runs in reverse: expected future aggregate sales are low relative to current sales, so $E_t[s_{t+1}/s_t]$ is small and the forward-looking term carries little weight. Pricing becomes more myopic, the current price absorbs more of the adjustment, and firms cut prices more aggressively, so the inflation response is amplified. The Euler equation now implies a more myopic marginal cost as well: with high γ_t , current marginal cost is tied primarily to current revenue from sales rather than to expected future wages, so mc_t falls by less on impact. This attenuates the price response but does not reverse it; the mispricing channel dominates.

Quantity channel. The asymmetric price responses feed into a quantity asymmetry through two general equilibrium forces. More deflation in the low-inventory state raises expected future deflation, which through the Fisher equation increases the ex-ante real interest rate; the household Euler equation (21) then implies a larger contraction in consumption and hence in sales. Separately, firms in the low-inventory state face a stronger motive to replenish stock and therefore produce more even as demand falls, widening the gap between production and sales and channeling the excess into inventory accumulation.

3.6 Model Limitations

The Lubik-Teo specification provides a tractable foundation for the mispricing channel, but several limitations are worth noting. First, the quantity asymmetry in the model is smaller than in the data. The model generates it primarily as a general equilibrium consequence of the differential price response (through the real rate and household Euler equation), rather than through a direct quantity channel. A richer account would require a mechanism by which low inventory directly amplifies the output decline — for instance through stockout avoidance motives, rationing, or search frictions in the product market. Second,

the mispricing channel operates entirely through the aggregate sales ratio s_{t+1}/s_t in the Phillips curve rather than through a deviation of an individual firm’s price from its optimal level: the Rotemberg cost penalizes the rate of price change scaled by aggregate sales, not the distance of the price from its target. A specification in which the cost of carrying inventory at a mispriced level rises with that distance — as in a menu-cost model — would capture the intuition more transparently. Third, the market clearing condition implies that at second order, consumption and sales diverge when prices adjust, which complicates the mapping of the model’s sales concept to the empirical sales data. We view these as avenues for future theoretical work and present the Lubik-Teo model as a first-pass illustration of the mispricing mechanism.

4 Conclusions

This paper establishes that the inventory position of firms is a first-order determinant of monetary policy transmission. A contractionary monetary policy shock contracts output and prices more than twice as much when firms enter the episode with low inventories as when they enter with high inventories; in the high-inventory regime the effects are small and statistically insignificant. We document this asymmetry at three levels of aggregation—an aggregate threshold VAR identified with external instruments, a panel of manufacturing and wholesale industries, and firm-level local projections that absorb sector-time fixed effects—and show that it survives a wide range of perturbations to the instrument, filter, sample, and variable specification. The firm-level evidence is particularly telling: because sector-time fixed effects absorb aggregate and sector-wide confounders, including previously documented financial channels, the inventory channel operates within narrowly defined industries rather than as a byproduct of the business cycle or of financial conditions.

These findings reframe a long-standing conclusion. Because production-smoothing models fail to match the procyclical volatility of production and the countercyclical inventory-

to-sales ratio, the literature has largely treated inventories as marginal to aggregate fluctuations. Our results show that, even if inventories do not amplify unconditional fluctuations, they profoundly shape the economy's conditional response to monetary policy. To rationalize the asymmetry, we embed inventories into an otherwise standard New Keynesian model with stock-elastic demand and price-adjustment costs proportional to sales. In this environment, high beginning-of-period inventories raise expected future sales and make firms' pricing decisions more forward-looking, dampening the response of inflation to a monetary shock; low inventories make pricing more myopic and the response larger. The model delivers the right qualitative asymmetry in both prices and quantities, and its limitations—most notably a quantity asymmetry smaller than in the data—point to natural directions for future theoretical work, such as stockout-avoidance motives, rationing, or search frictions that would let low inventories amplify output responses directly.

The results carry implications for the conduct of monetary policy. The strength of the transmission mechanism is not constant but varies systematically with an observable state variable, the inventory-to-sales ratio. A given policy action will be considerably more contractionary when aggregate inventories are lean than when they are abundant, suggesting that policymakers may wish to condition the scale and timing of their interventions on the inventory cycle. More broadly, our evidence adds the inventory position to the growing list of real-side state variables—alongside capacity utilization and supply-chain conditions—that govern the slope of the aggregate supply curve and, with it, the trade-off central banks face between stabilizing output and stabilizing prices.

References

- AHLANDER, E., N. AMBERG, AND M. KLEIN (2026): “Do Capacity Constraints Affect the Pass-Through of Monetary Policy to Prices?” Tech. rep., Sveriges Riksbank, working paper.
- ALESSANDRI, P. AND H. MUMTAZ (2017): “Financial conditions and density forecasts for US output and inflation,” *Review of Economic Dynamics*, 24, 66–78.
- (2019): “Financial regimes and uncertainty shocks,” *Journal of Monetary Economics*, 101, 31–46.
- ARUOBA, S. B. AND T. DRECHSEL (2024): “Identifying Monetary Policy Shocks: A Natural Language Approach,” *American Economic Journal: Macroeconomics*, forthcoming.
- BAI, X., J. FERNÁNDEZ-VILLAYERDE, Y. LI, AND F. ZANETTI (2026): “State Dependence of Monetary Policy During Global Supply Chain Disruptions,” Tech. rep., working paper.
- BAUER, M. D. AND E. T. SWANSON (2023): “A Reassessment of Monetary Policy Surprises and High-Frequency Identification,” *NBER Macroeconomics Annual*, 37, 87–155.
- BERNANKE, B. S. AND M. GERTLER (1995): “Inside the Black Box: The Credit Channel of Monetary Policy Transmission,” *Journal of Economic Perspectives*, 9, 27–48.
- BILS, M. AND J. A. KAHN (2000): “What Inventory Behavior Tells Us About Business Cycles,” *American Economic Review*, 90, 458–481.
- BLINDER, A. S. (1982): “Inventories and Sticky Prices: More on the Microfoundations of Macroeconomics,” *American Economic Review*, 72, 334–348.
- BLINDER, A. S. AND L. J. MACCINI (1991): “Taking Stock: A Critical Assessment of Recent Research on Inventories,” *Journal of Economic Perspectives*, 5, 73–96.
- BOARD OF GOVERNORS OF THE FEDERAL RESERVE SYSTEM (2023): “Industrial Production and Capacity Utilization – G.17,” <https://www.federalreserve.gov/releases/g17/>.
- BOEHM, C. E. AND N. PANDALAI-NAYAR (2022): “Convex Supply Curves,” *American Economic Review*, 112, 3941–3969.
- CLOYNE, J., C. FERREIRA, M. FROEMEL, AND P. SURICO (2023): “Monetary Policy, Corporate Finance, and Investment,” *Journal of the European Economic Association*, 21, 2586–2634.
- DE SOUSA RODRIGUES, D. AND T. WILLEMS (2024): “Inventories Matter for the Transmission of Monetary Policy: Uncovering the Cost-of-Carry Channel,” Tech. rep., Bank of England, working paper.
- DEGASPERI, R. AND G. RICCO (2024): “The Identification of Monetary Policy Shocks,” *Review of Economic Studies*.

- DEN HAAN, W. J. (2014): “Inventories and the role of goods-market frictions for business cycles,” Tech. rep.
- FAVARA, G., S. GILCHRIST, K. F. LEWIS, AND E. ZAKRAJŠEK (2016): “Updating the Recession Risk and the Excess Bond Premium,” FEDS notes, Board of Governors of the Federal Reserve System, october 6, 2016.
- FRANCIS, N., M. OWYANG, AND D. F. SOQUES (2023): “Impulse Response Functions for Self-Exciting Nonlinear Models,” Tech. rep., National Bureau of Economic Research.
- GERTLER, M. AND S. GILCHRIST (1994): “Monetary Policy, Business Cycles, and the Behavior of Small Manufacturing Firms,” *Quarterly Journal of Economics*, 109, 309–340.
- GILCHRIST, S. AND E. ZAKRAJŠEK (2012): “Credit Spreads and Business Cycle Fluctuations,” *American Economic Review*, 102, 1692–1720.
- GONÇALVES, S., A. M. HERRERA, L. KILIAN, AND E. PESAVENTO (2024): “State-dependent local projections,” *Journal of Econometrics*, 244, 105702.
- IRELAND, P. N. (2004): “Technology shocks in the new Keynesian model,” *Review of Economics and Statistics*, 86, 923–936.
- JEENAS, P. (2024): “Firm Balance Sheet Liquidity, Monetary Policy Shocks, and Investment Dynamics,” *Journal of Political Economy*, forthcoming.
- JORDÀ, Ò. (2005): “Estimation and inference of impulse responses by local projections,” *American economic review*, 95, 161–182.
- KASHYAP, A. K., O. A. LAMONT, AND J. C. STEIN (1994): “Credit Conditions and the Cyclical Behavior of Inventories,” *Quarterly Journal of Economics*, 109, 565–592.
- KHAN, A. AND J. K. THOMAS (2007): “Inventories and the Business Cycle: An Equilibrium Analysis of (S,s) Policies,” *American Economic Review*, 97, 1165–1188.
- KOOP, G., M. H. PESARAN, AND S. M. POTTER (1996): “Impulse response analysis in nonlinear multivariate models,” *Journal of Econometrics*, 74, 119–147.
- LUBIK, T. A. AND W. L. TEO (2012): “Inventories, Inflation Dynamics, and the New Keynesian Phillips Curve,” *European Economic Review*, 56, 327–346.
- MCCRACKEN, M. W. AND S. NG (2016): “FRED-MD: A monthly database for macroeconomic research,” *Journal of Business & Economic Statistics*, 34, 574–589.
- MEHROTRA, N., H. OH, AND J. ORTIZ (2025): “Retail Inventories and Inflation Dynamics: The Price Margin Channel,” Tech. rep., Federal Reserve Board, working paper.
- MIRANDA-AGRIPPINO, S. AND G. RICCO (2021): “The Transmission of Monetary Policy Shocks,” *American Economic Review: Insights*, 3, 369–387.

- MONTIEL OLEA, J. L. AND M. PLAGBORG-MØLLER (2021): “Local projection inference is simpler and more robust than you think,” *Econometrica*, 89, 1789–1823.
- OTTONELLO, P. AND T. WINBERRY (2020): “Financial Heterogeneity and the Investment Channel of Monetary Policy,” *Econometrica*, 88, 2473–2502.
- PLAGBORG-MØLLER, M. AND C. K. WOLF (2021): “Local projections and VARs estimate the same impulse responses,” *Econometrica*, 89, 955–980.
- ROMER, C. D. AND D. H. ROMER (2004): “A new measure of monetary shocks: Derivation and implications,” *American Economic Review*, 94, 1055–1084.
- STOCK, J. H. AND M. W. WATSON (2018): “Identification and Estimation of Dynamic Causal Effects in Macroeconomics Using External Instruments,” *The Economic Journal*, 128, 917–948.
- U.S. BUREAU OF LABOR STATISTICS (2023): “Producer Price Indexes (PPI): Industry Data,” <https://www.bls.gov/ppi/>.
- U.S. CENSUS BUREAU (2023a): “Advance Monthly Manufacturers’ Shipments, Inventories, and Orders (M3),” <https://www.census.gov/manufacturing/m3/index.html>.
- (2023b): “Monthly Retail Trade Survey (MRTS),” <https://www.census.gov/retail/index.html>.
- (2023c): “Monthly Wholesale Trade Survey (MWTS),” <https://www.census.gov/wholesale/index.html>.
- WIELAND, J. (2021): “Updated Romer-Romer Monetary Policy Shocks,” Tech. rep., Ann Arbor, MI: Inter-university Consortium for Political and Social Research, <https://doi.org/10.3886/E135741V1>.

A Data

A.1 Aggregate data

The table below lists the aggregate variables used throughout the paper.

1-Year treasury bill rate *Source:* FRED-MD Monthly Macroeconomic Database; *Code:* GS1; *Reference:* [McCracken and Ng \(2016\)](#); *Transformation:* Percent per annum; normalized to basis points in figures.

Industrial production *Source:* FRED-MD Monthly Macroeconomic Database; *Code:* IN-DPRO; *Reference:* [McCracken and Ng \(2016\)](#); *Transformation:* Logarithmic.

Producer price index *Source:* FRED-MD Monthly Macroeconomic Database; *Code:* PPIFGS (Producer price index: finished goods); *Reference:* [McCracken and Ng \(2016\)](#); *Transformation:* Logarithmic.

Commodity prices *Source:* FRED-MD Monthly Macroeconomic Database; *Code:* CUSR0000SAC (CPI for all urban consumers: commodities); *Reference:* [McCracken and Ng \(2016\)](#); *Transformation:* Logarithmic.

Excess bond premium *Source:* Federal Reserve Board website; *Code:* EBP; *Reference:* [Gilchrist and Zakrajšek \(2012\)](#) and extended by [Favara et al. \(2016\)](#); *Transformation:* Percent; normalized to basis points in figures.

Finished-goods inventory-to-sales ratio *Notes:* Ratio of two objects. *Transformation:* Band-pass filtered cyclical component (periodicities 2–120 months). **Numerator** Finished-goods inventories is the sum of finished-goods inventories of the manufacturing sector, total inventories of the wholesale sector, and total inventories of the retail sector, drawn respectively from the Census Bureau’s M3 ([U.S. Census Bureau, 2023a](#)), MWTS ([U.S. Census Bureau, 2023c](#)), and MRTS ([U.S. Census Bureau, 2023b](#)) surveys (all seasonally adjusted).⁸ **Denominator** Current-quarter total sales are defined as total business sales,

⁸ The Census Bureau redesigned these surveys from the Standard Industrial Classification (SIC) to the North American Industry Classification System (NAICS) in the early 2000s, so NAICS-based data are available only

recovered from [McCracken and Ng \(2016\)](#) as $BUSINV_x_t / ISRATIO_x_t$, computed as the three-month cumulative sum $s_{t-2} + s_{t-1} + s_t$.

Consumer price index *Source:* FRED-MD Monthly Macroeconomic Database; *Code:* CPI-AUCSL (CPI for all urban consumers: all items); *Reference:* [McCracken and Ng \(2016\)](#); *Transformation:* Logarithmic.

Backlogs of orders *Source:* US Census Bureau, Monthly Survey of Manufacturers (M3); *Reference:* [U.S. Census Bureau \(2023a\)](#); *Coverage:* Manufacturing sector only; *Transformation:* In the robustness check, backlogs of orders are subtracted from finished-goods manufacturing inventories in the numerator of the inventory-to-sales ratio as described above; the denominator remains unchanged.

Unemployment rate *Source:* FRED-MD Monthly Macroeconomic Database; *Code:* UNRATE (civilian unemployment rate); *Reference:* [McCracken and Ng \(2016\)](#); *Transformation:* Percentage points.

A.2 Sectoral data

The sectoral dataset is an unbalanced panel of 35 three-digit NAICS industries observed at monthly frequency from January 1992 to December 2023. The panel is unbalanced for two reasons: (a) industrial production and order-based series (e.g., unfilled orders) are not defined for wholesale and retail industries because firms operating in these industries do not produce goods but promptly distribute them; (b) PPI coverage varies by sector and begins after January 1992 for many industries. All quantity and inventory series are seasonally adjusted. PPI series are not seasonally adjusted, as the BLS does not publish seasonally adjusted PPI at the three-digit NAICS level.

Manufacturers' Shipments, Inventories, and Orders (M3). *Source:* U.S. Census Bureau; *Reference:* [U.S. Census Bureau \(2023a\)](#); *Coverage:* 21 manufacturing industries (NAICS 311–339), January 1992 – December 2023. The M3 survey provides, for each

from that point onward. In order to have data since December 1980, we extend the series by matching the corresponding historical SIC-based data to the NAICS-based series.

sector: value of shipments (SALES), new orders (NewOrders), unfilled orders (UnfilledOrders), total inventories (TotInv), finished-goods inventories (OutInv), materials and supplies inventories (MatInv), work-in-process inventories (WIPInv), and the inventory-to-sales ratio (ISRatio). All series are seasonally adjusted.

Monthly Wholesale Trade Survey (MWTS). *Source:* U.S. Census Bureau; *Reference:* [U.S. Census Bureau \(2023c\)](#); *Coverage:* 2 wholesale industries (NAICS 423, 424), January 1992 – December 2023. The MWTS provides monthly sales, total inventories, and the inventory-to-sales ratio. No sub-component breakdown of inventories (finished goods, materials, WIP) is available for wholesale; accordingly, TotInv equals OutInv for these sectors, and NewOrders, UnfilledOrders, MatInv, and WIPInv are not collected.

Monthly Retail Trade Survey (MRTS). *Source:* U.S. Census Bureau; *Reference:* [U.S. Census Bureau \(2023b\)](#); *Coverage:* 12 retail industries (NAICS 441–454), January 1992 – ongoing. The MRTS provides monthly sales, total inventories, and the inventory-to-sales ratio. As with wholesale, TotInv equals OutInv, and order-based and input-inventory series are not available.

Industrial Production and Capacity Utilization (G.17). *Source:* Board of Governors of the Federal Reserve System; *Reference:* [Board of Governors of the Federal Reserve System \(2023\)](#); *Coverage:* 21 manufacturing industries (NAICS 311–339), January 1992 – ongoing. The G.17 release provides monthly industrial production indices (IP) and capacity utilization indices (CAP) for each manufacturing sector. These series are not available for wholesale or retail industries.

Producer Price Index, Industry Data. *Source:* U.S. Bureau of Labor Statistics; *Reference:* [U.S. Bureau of Labor Statistics \(2023\)](#); *Coverage:* 20 manufacturing, 2 wholesale, and 1 retail industry (23 industries total). Series are not seasonally adjusted. The BLS publishes these three-digit NAICS aggregate series from different start dates: for 8 manufacturing sectors (NAICS 311, 324, 325, 326, 327, 331, 332, 337), coverage begins in January 1992;

for the remaining 12 manufacturing sectors, coverage begins in December 2003; for the 2 wholesale sectors (NAICS 423 and 424), coverage begins in June 2004 and June 2005, respectively; for NAICS 441, coverage begins in December 2003. NAICS 316 (Leather and Allied Product Manufacturing) and retail sectors NAICS 442–454 have no published three-digit series; PPI is set to NaN for these sectors throughout. To ensure comparability across months within each sector, PPI is sourced exclusively from the BLS three-digit aggregate series; no mixing of methods is applied.

Table 2 lists all 35 industries, their data sources by variable group, and the PPI coverage start date.

Table 2: Sectoral panel dataset: industry coverage and variable sources

NAICS	Description	IP	Inv./Sales	PPI	PPI start
<i>Panel A: Manufacturing (21 sectors) — M3 + G.17, January 1992 – December 2023</i>					
311	Food Manufacturing	G.17	M3	BLS	Jan 1992
312	Beverage and Tobacco Product Mfg	G.17	M3	BLS	Dec 2003
313	Textile Mills	G.17	M3	BLS	Dec 2003
314	Textile Product Mills	G.17	M3	BLS	Dec 2003
315	Apparel Manufacturing	G.17	M3	BLS	Dec 2003
316	Leather and Allied Product Mfg	G.17	M3	—	—
321	Wood Product Manufacturing	G.17	M3	BLS	Dec 2003
322	Paper Manufacturing	G.17	M3	BLS	Dec 2003
323	Printing and Related Support Activities	G.17	M3	BLS	Dec 2003
324	Petroleum and Coal Products Mfg	G.17	M3	BLS	Jan 1992
325	Chemical Manufacturing	G.17	M3	BLS	Jan 1992
326	Plastics and Rubber Products Mfg	G.17	M3	BLS	Jan 1992
327	Nonmetallic Mineral Product Mfg	G.17	M3	BLS	Jan 1992
331	Primary Metal Manufacturing	G.17	M3	BLS	Jan 1992
332	Fabricated Metal Product Mfg	G.17	M3	BLS	Jan 1992
333	Machinery Manufacturing	G.17	M3	BLS	Dec 2003
334	Computer and Electronic Product Mfg	G.17	M3	BLS	Dec 2003
335	Electrical Equipment, Appliance & Component	G.17	M3	BLS	Dec 2003
336	Transportation Equipment Mfg	G.17	M3	BLS	Dec 2003
337	Furniture and Related Product Mfg	G.17	M3	BLS	Jan 1992
339	Miscellaneous Manufacturing	G.17	M3	BLS	Dec 2003
<i>Panel B: Wholesale (2 sectors) — MWTS, January 1992 – December 2023</i>					
423	Merchant Wholesalers, Durable Goods	—	MWTS	BLS	Jun 2004
424	Merchant Wholesalers, Nondurable Goods	—	MWTS	BLS	Jun 2005
<i>Panel C: Retail (12 sectors) — MRTS, January 1992 – December 2023</i>					
441	Motor Vehicle and Parts Dealers	—	MRTS	BLS	Dec 2003
442	Furniture and Home Furnishings Stores	—	MRTS	—	—
443	Electronics and Appliance Stores	—	MRTS	—	—
444	Building Material and Garden Supply Dealers	—	MRTS	—	—
445	Food and Beverage Stores	—	MRTS	—	—
446	Health and Personal Care Stores	—	MRTS	—	—
447	Gasoline Stations	—	MRTS	—	—
448	Clothing and Clothing Accessories Stores	—	MRTS	—	—
451	Sporting Goods, Hobby, Book & Music Stores	—	MRTS	—	—
452	General Merchandise Stores	—	MRTS	—	—
453	Miscellaneous Store Retailers	—	MRTS	—	—
454	Nonstore Retailers	—	MRTS	—	—

Notes: IP and CAP (G.17) are available for manufacturing only. For wholesale and retail, TotInv = OutInv (no input-inventory breakdown); NewOrders, UnfilledOrders, MatInv, and WIPInv are not collected. PPI “—” denotes no published BLS three-digit NAICS series.

Balanced subset. TO BE ADJUSTED For the local projection in Section 2.2, we restrict attention to the 20 manufacturing sectors with all four core variables simultaneously non-missing: industrial production (IP), value of shipments (SALES), total inventories (TotInv), and producer price index (PPI). This excludes NAICS 316, which has no published BLS three-digit PPI series. The 20 sectors fall into two groups by PPI availability. *Group A* (8 sectors: NAICS 311, 324, 325, 326, 327, 331, 332, 337) has PPI coverage from January 1992, yielding 382 monthly observations per sector over January 1992 – October 2023. *Group B* (12 sectors: NAICS 312, 313, 314, 315, 321, 322, 323, 333, 334, 335, 336, 339) has PPI coverage from December 2003, yielding 239 monthly observations per sector over December 2003 – October 2023. The common balanced range over which all 20 sectors have complete data is December 2003 – October 2023 (239 months, 4,780 observations).